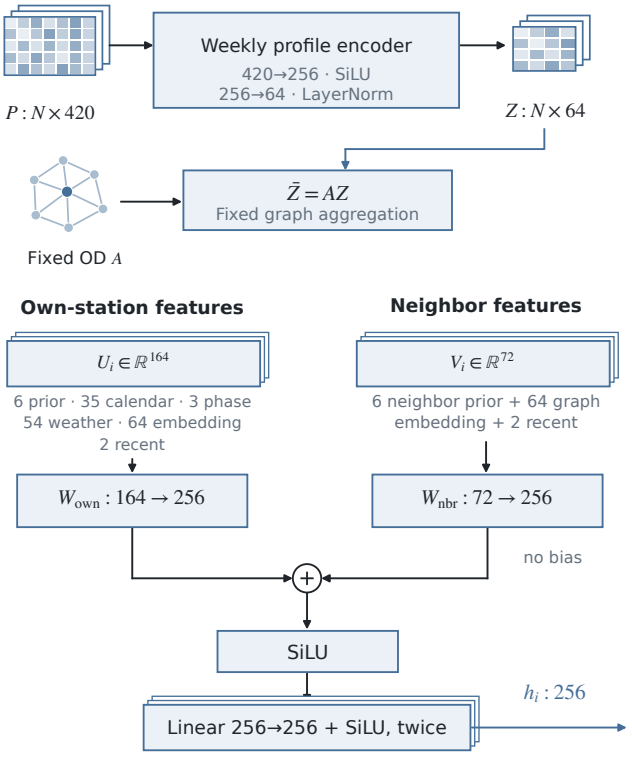
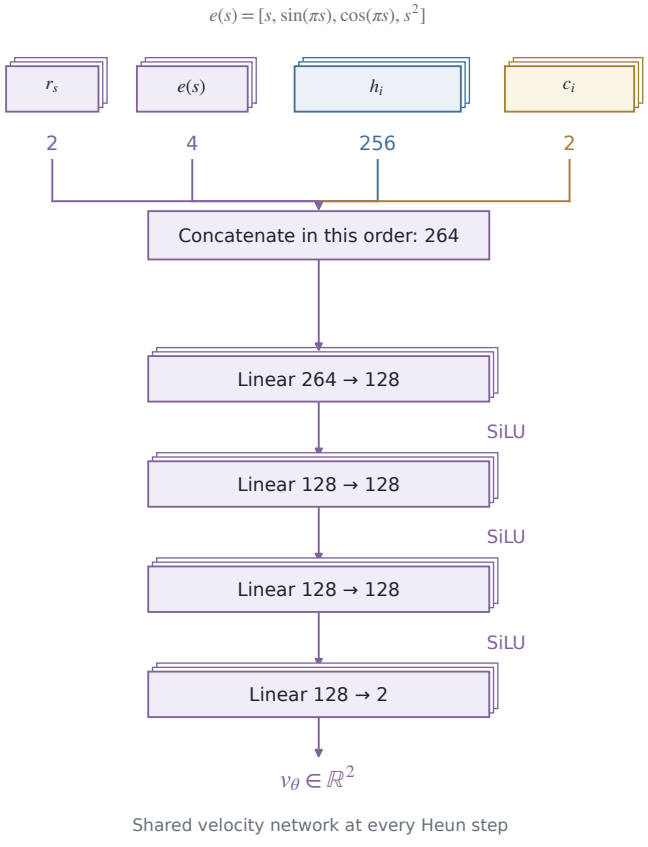
(a) Encoder: graph and context fusion



(b) Decoder: conditional velocity network



Shared velocity network at every Heun step

(c) Two-stage historical learning; fixed-weight API evaluation

$$r_0 \sim \mathcal{N}(0, I_2), \qquad r_1 = \log(1 + y + U) - \log(1 + \lambda), \qquad U \sim \mathcal{U}(0, 1)$$

$$r_s = (1 - s)r_0 + sr_1, \qquad \mathcal{L}_{\text{CFM}} = \mathbb{E}\|v_\theta(r_s, s, h, c) - (r_1 - r_0)\|^2$$

Rate network fitted first and frozen for FM. API observations enter evaluation / starting state; no gradient returns to either network.